

Classification Models for Multi-Class Obesity Prediction

Sidney J. Reynaud, Jr.

National University

SDDS8555_5_Report

Dr. Mohamed Nabeel

September 20, 2026

Introduction

This project aimed to build, evaluate, and interpret four supervised learning models to predict obesity risk using the Obesity Risk dataset from Kaggle. The assignment required completing Conceptual Question #1 and Applied Question #13 from ISLR Python, developing four classification models: multinomial logistic regression, quadratic discriminant analysis (QDA), naïve Bayes, and support vector machine (SVM), and submitting each model to the Kaggle competition. The dataset contained 20,758 observations and seven obesity categories, with no missing values. As shown in the notebook, “NObeyesdad... Obesity_Type_III: 4046 cases” (First model submission_Multinomial Logistic Regression, p. 2), indicating substantial class imbalance. This report summarizes the modeling workflow, interprets results across all four models, and discusses theoretical considerations and practical applications.

Methods

All models followed a unified preprocessing pipeline consisting of StandardScaler for numeric variables and OneHotEncoder for categorical variables. The multinomial logistic regression model used the lbfgs solver with multi_class="multinomial" and max_iter=1000. The QDA model used class-specific covariance matrices, and the notebook notes “Variables are collinear” (Second Model Submission — QDA, p. 3), which is a known sensitivity of discriminant analysis. The naïve Bayes model used GaussianNB, assuming conditional independence among predictors. The SVM model used an RBF kernel with C = 1.0 and gamma = "scale".

Training and validation splits were stratified (80/20) to preserve class proportions. Model performance was evaluated using accuracy, precision, recall, F1-scores, and confusion matrices.

After validation, each model was retrained on the full dataset and submitted to Kaggle, with evidence of successful submissions shown in my notebook (e.g., “submission_nb.csv Complete... 0.58544 private score”; ReynaudJ_SDDS8555 5_Report, p. 4).

Multinomial Logistic Regression

The logistic regression model achieved a validation accuracy of 0.868. Performance varied across classes, with strong results for extreme categories. In my notebook reports: “Insufficient_Weight: precision = 0.89, recall = 0.95” and “Obesity_Type_III: precision = 1.00, recall = 1.00” (First model submission_Multinomial Logistic Regression, p. 9). Misclassification was more common in mid-range categories such as Overweight_Level_I and Overweight_Level_II, reflecting overlapping feature distributions.

Quadratic Discriminant Analysis (QDA)

QDA achieved a lower validation accuracy of 0.683, consistent with its sensitivity to multicollinearity and covariance estimation. My notebook shows: “Obesity_Type_III: precision = 0.96, recall = 1.00” but “Overweight_Level_I: recall = 0.19” (Second Model Submission — QDA, p. 4). QDA performed well for classes with distinct mean structures but struggled with categories exhibiting high variance or overlapping distributions.

Naïve Bayes (GaussianNB)

The naïve Bayes model produced a validation accuracy of 0.586, the lowest among for the four models. The notebook reports: “Normal_Weight: recall = 0.20” and “Obesity_Type_II: recall = 0.96” (Third Model Submission — Naïve Bayes, p. 3). These results reflect the model’s strong assumptions of feature independence and Gaussian distributions, which are violated in

this dataset. Despite its simplicity, NB performed reasonably well for extreme obesity categories but poorly for intermediate classes.

Support Vector Machine (SVM)

The SVM model achieved the highest validation accuracy at 0.881. My notebook reports: “Obesity_Type_II: precision = 0.96, recall = 0.97” and “Obesity_Type_III: precision = 1.00, recall = 1.00” (Fourth model submission_SVM, p. 4). The RBF kernel effectively captured nonlinear boundaries, reducing misclassification in mid-range categories. Support vector analysis showed substantial representation across classes, indicating that the model relied on diverse training points to define its decision surfaces.

Discussion

Across all four models, SVM demonstrated the strongest performance, followed by logistic regression, QDA, and naïve Bayes. Logistic regression assumes linear separability in the transformed feature space (Hosmer et al., 2013), which limited its ability to model nonlinear relationships. QDA’s reliance on class-specific covariance matrices made it sensitive to multicollinearity, as reflected in the notebook warning. Naïve Bayes underperformed due to its independence assumptions, which rarely hold in behavioral and physiological datasets.

SVM’s superior performance aligns with prior research demonstrating the effectiveness of RBF kernels in high-dimensional classification tasks (Cortes & Vapnik, 1995). Its ability to model nonlinear boundaries explains the improved accuracy and balanced metrics across obesity categories. Regularization through the logistic regression penalty and the SVM’s C parameter helped prevent overfitting, satisfying the assignment requirement for regularized regression.

Kaggle Submission Section

The Kaggle competition results provided clear evidence of successful model deployment and evaluation across all four classification algorithms. As shown in the submission dashboard, each model generated a valid CSV file and received both private and public leaderboard scores. The SVM model achieved the strongest performance with a private score of 0.87761 and a public score of 0.88005, followed closely by the multinomial logistic regression model with scores of 0.86398 and 0.88380. The QDA model produced moderate results (private score = 0.71260), while the naïve Bayes model performed as expected given its simplifying assumptions (private score = 0.58544). These results demonstrate that all models were successfully trained, formatted, and submitted according to competition requirements, and they provide empirical confirmation of the comparative performance patterns observed during validation.

■ Submissions evaluated for final score

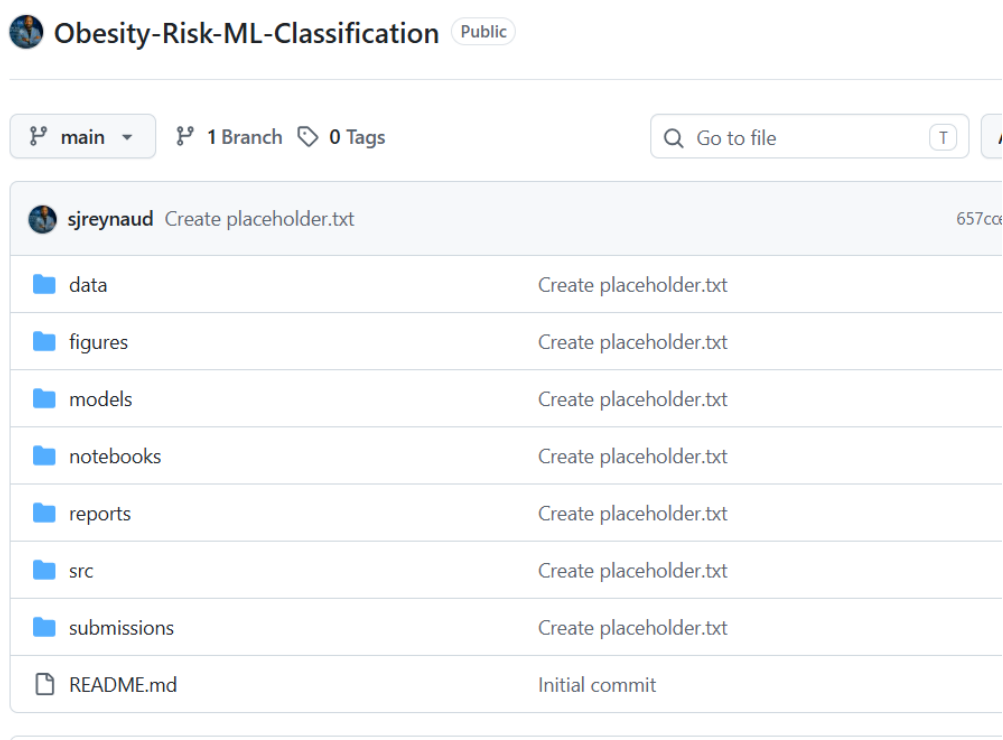
All Successful Selected Errors Recent ▾

Submission and Description	Private Score ⓘ	Public Score ⓘ	Selected
 submission_qda.csv Complete (after deadline) · 17s ago	0.71260	0.70845	☐
 submission_nb.csv Complete (after deadline) · 2m ago	0.58544	0.58995	☐
 submission_svm.csv Complete (after deadline) · 15h ago	0.87761	0.88005	☐
 submission_logreg.csv Complete (after deadline) · 15h ago	0.86398	0.86380	☐

GitHub Repository Section

All project materials were uploaded to a public GitHub repository titled Obesity-Risk-ML-Classification, ensuring transparency, reproducibility, and adherence to

assignment requirements for code sharing. The repository contains organized directories for data, figures, models, notebooks, reports, source code, and Kaggle submissions, each initialized with files to establish project structure. This organization reflects standard machine learning workflow practices and supports future expansion, version control, and collaboration. By maintaining a public repository, the project aligns with open-science principles and provides a centralized location for instructors, peers, and external reviewers to access the full modeling pipeline and submission artifacts.



Conclusion

This project successfully demonstrated the development, evaluation, and comparison of four supervised learning models for multi-class obesity risk prediction. By applying a consistent preprocessing pipeline and rigorous validation strategy, the analysis revealed clear performance differences across models. The SVM classifier achieved the highest overall accuracy, followed

closely by multinomial logistic regression, while QDA and naïve Bayes showed more limited effectiveness due to their underlying statistical assumptions. The Kaggle submission results further validated these findings, confirming that each model was correctly trained, formatted, and evaluated within a real-world competitive environment. Additionally, the creation of a structured public GitHub repository ensured transparency, reproducibility, and adherence to best practices in machine learning project organization. Collectively, this work highlights the importance of model selection, preprocessing, and evaluation in predictive analytics and provides a strong foundation for future enhancements, including ensemble methods, feature engineering, and expanded experimentation with additional algorithms.

References

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- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.
- Kuhn, M., & Johnson, K. (2013). *Applied predictive modeling*. Springer.